

KothaKhoj.com

Project Journal & Reference Manual

The complete story of building and launching a rental-room map for Butwal, Nepal - from local errors to a live website. Built by Satish Mallah Sahani on the open-source Shareabouts platform (GPL v3, by OpenPlans).

Live site: <https://kothakhaj.com> **API:** <https://api.kothakhaj.com>

Journal covers: July 12 - 18, 2026 (local development to deployment)

1. What the System Is

KothaKhoj is two Django applications working together, plus supporting services, all running in Docker containers on a single DigitalOcean server in Bangalore:

The Map app (folder: shareabouts) - what visitors see. Shows the Leaflet map, place forms, search box, directions. It has no database of its own; it proxies all data requests to the API, attaching a secret dataset key.

The API app (folder: shareabouts-api) - stores everything: places, comments, support votes, users. PostGIS database + Redis + Celery worker behind it. Admin panel at api.kothakhoy.com/admin (user: admin).

Traffic path: Visitor -> Cloudflare (free CDN, HTTPS, Kathmandu edge) -> Caddy on the server (automatic HTTPS certificates) -> map app (port 8080) or API (port 8000).

The flavor: all customization lives in `src/flavors/satish/` - `config.yml` controls place types (single_room, double_room, flat), form fields, map layers, and texts.

2. Infrastructure Facts

Thing	Value / Where
Domain	kothakhoy.com (Namecheap, ~\$7/yr, auto-renew ON)
DNS + CDN	Cloudflare free plan; A records @ and api -> droplet IP; SSL Full (strict)
Server	DigitalOcean droplet, Bangalore, Ubuntu 24.04, \$12/mo (1 CPU / 2 GB / 50 GB)
Server IP	165.232.180.13 (SSH: ssh root@165.232.180.13)
HTTPS	Caddy (/etc/caddy/Caddyfile) - automatic certificates
Firewall	ufw: only ports 22, 80, 443 open; DB/Redis bound to localhost
Map tiles/search	Mapbox - production token restricted to kothakhoy.com URLs
College search data	Google Sheet published as CSV (edit sheet = update search)

Secrets (never in git, never share): server root password, Django admin password, two SECRET_KEYS, Mapbox tokens. Locations: map settings in `/opt/shareabouts/src/project/local_settings.py`; API settings in `/opt/shareabouts-api/.env`. Same files exist on the Mac for local development (with localhost values).

3. Problems We Solved (and What They Taught)

404 when adding a place

Symptom: every `/api/places` call returned 404. **Cause:** `DATASET_ROOT` in `local_settings.py` pointed to a dataset (owner/slug) that did not exist on the API. **Fix:** point it at the real dataset (owner `demo-user`, slug `demo-data`) with the matching `DATASET_KEY`. **Lesson:** the dataset URL format is `/api/v2/OWNER/datasets/SLUG` - owner is the API user who owns the dataset, not just any admin login.

'Place type undefined' console warning

Harmless: fires for a brand-new unsaved place before a type is chosen, because the type-check runs before the `isNew()` check in `layer-view.js`. Ignore it.

Flat (orange circle) markers not clickable

Cause: flats were the only type drawn as vector circles; circles live in a lower map layer than icon markers, and the Butwal boundary polygon was accidentally click-blocking (old Leaflet 'clickable' option is ignored by Leaflet 1.x - should be 'interactive'). **Fix:** gave flats icon markers (`dot-f95016.png` / `marker-f95016.png`) like other types. **Lesson:** icons always win clicks over drawn shapes.

Safari: directions / location silently failing

Cause: Safari only allows geolocation on HTTPS pages - and unlike Chrome it makes NO exception for 127.0.0.1. It fails silently: no prompt, no error. **Fix in development:** test location features in Chrome. **Fix forever:** the live site is HTTPS, so everything works everywhere now.

CSRF 403 on second comment

Cause: map app and API are both Django on the same address locally - they fight over the `csrftoken` cookie; each successful comment poisons the next request. **Status:** left unfixed by choice (workaround: refresh). If it ever annoys real users, the fix is stripping the API's `csrftoken` cookie in the map proxy (`sa_web/views.py api()`).

Delete button needing a page refresh sometimes

Cause: timing race - the panel re-renders about a second after adding a place (when comment counts arrive); a click during the redraw dies. Rare, self-healing, left as is.

init container failed on server (exit 1)

Cause: invalid character/stray edit in `DJANGO_SUPERUSER_USERNAME` line of `.env`. **Lesson:** in `.env` files avoid `#`, `$`, quotes and spaces in values; long letters+numbers passwords are safest and strongest.

400 Bad Request when curling the API on the server

Not an error! `ALLOWED_HOSTS` protection working - Django refuses requests addressed to the wrong hostname. Test with: `curl -H "Host: api.kothakhoj.com" http://localhost:8000/...`

4. Features Built On Top of Stock Shareabouts

Custom flavor 'satish': three place types with icons, Nepali-context form (contact number optional + validated pattern 98/97/96, labels on detail view), Butwal map boundary, simplified survey (comment + name only).

Local search box: merged search - Google Sheet place names (with Nepali aliases, fuzzy matching via Fuse.js) + Mapbox geocoding, limited to the Butwal region. 92 Rupandehi colleges/schools imported with coordinates from OpenStreetMap.

Delete own place: button appears only for the creator (user_token match), with confirmation. PLUS a server-side ownership check in the proxy (sa_web/views.py) - forged delete requests from other sessions get 403 Forbidden. Verified with curl attack tests.

Live directions: DIRECTIONS button on every place - Leaflet Routing Machine + Mapbox walking directions from the visitor's live GPS position, re-routing as they move (15 m threshold, accuracy filter). Turn-by-turn panel, auto-cleanup when panel closes.

Locate button: centers the map exactly on the visitor's position (high-accuracy GPS) so the movable add-place pin starts right where they stand.

5. Routines You Will Use Again and Again

Daily development loop (on the Mac)

1. edit code
2. test at <http://127.0.0.1:8080> (Chrome for location features)
3. `cd ~/shareabouts && git add -A && git commit -m "what changed"`

Git safety net

```
git status          # what changed?
git diff            # exact changed lines
git log --oneline   # list of save points
git restore FILE    # undo one file to last commit
git reset --hard CODE # jump whole project to a save point
rm ~/shareabouts/.git/index.lock # if git complains about a lock
```

Deploying an update (map app)

```
# on Mac - upload changed code:
rsync -av --exclude env --exclude .git --exclude node_modules \
  ~/shareabouts root@165.232.180.13:/opt/

# on server (ssh root@165.232.180.13) - rebuild and restart:
cd /opt/shareabouts && \
docker build --build-arg SHAREABOUTS_FLAVOR=satish -t kothakhoj-map . && \
docker rm -f kothakhoj-map && \
docker run -d --restart always --name kothakhoj-map \
  -p 127.0.0.1:8080:8000 kothakhoj-map
```

Deploying an update (API app)

```
rsync -av --exclude env --exclude .git ~/shareabouts-api root@165.232.180.13:/opt/
# then on server:
cd /opt/shareabouts-api && docker compose up -d --build
```

Server health checks

```
docker ps          # everything running?
cd /opt/shareabouts-api && docker compose ps
docker logs kothakhoj-map --tail 50 # map app logs
docker compose logs web --tail 50   # API logs
systemctl status caddy              # HTTPS server
```

Admin operations (on server)

```
# change admin password:
cd /opt/shareabouts-api && docker compose exec web \
  python3 manage.py changepassword admin
# list all places / delete by id:
docker compose exec web python3 manage.py shell -c \
  "from sa_api_v2.models import Place;\n  [print(p.id, str(p.data)[:80]) for p in Place.objects.all()]"
# admin panel in browser: https://api.kothakhoj.com/admin
```

6. Security Posture (as of launch)

Done: DEBUG off on both apps - unique SECRET_KEYS - strong admin password - ALLOWED_HOSTS locked to real domains - database/Redis/app ports bound to localhost only - ufw firewall (22/80/443) - HTTPS everywhere via Caddy + Cloudflare Full (strict) - server-side delete ownership check - Mapbox production token URL-restricted - domain privacy on - secrets excluded from git.

Known remaining risks (accepted for now): anonymous posting with no rate limit (spam possible - planned fix: Facebook login requirement) - map app Django 3.2 is past end-of-life (upgrade to 4.2 someday) - old jQuery/Handlebars libraries - CSRF cookie clash papercut - root SSH login by password (consider SSH keys later).

7. To-Do List (in rough priority order)

- 1. Branding fix:** site still titled 'Bhada Ghar' - change title/meta_description in src/flavors/satish/config.yml app section to KothaKhoj + good Nepali/English description, then redeploy. Do this BEFORE requesting Google indexing.
- 2. Google Search Console:** add property kothakhaj.com, verify via Cloudflare TXT record, request indexing. Expect search visibility in 1-2 weeks.
- 3. Marketing:** share in Butwal Facebook groups; every real link helps ranking.
- 4. Backups:** enable DigitalOcean droplet backups (~\$2.40/mo) once real users add data.
- 5. GitHub Student Pack:** if eligible - \$200 DigitalOcean credit (education.github.com/pack).
- 6. Facebook login** (needs the live domain - now possible): FB developer app, configure social auth in API, remove Twitter button, require login to post. Also fixes spam risk.
- 7. Phone number OTP verification:** Sparrow SMS account + two small endpoints + form JS.
- 8. Mapbox map tiles** for smoother panning (config.yml layers change).
- 9. Django 4.2 upgrade** for the map app (bigger task, schedule calmly).

8. Big Lessons From This Project

The browser lies, the server decides. Hiding a button is cosmetics; real rules are enforced server-side (delete ownership, ALLOWED_HOSTS, login).

Git is a save button. Commit after every working step; every mess becomes reversible.

Change one thing, test, commit. Every bug found this week was found fast because only one thing had changed.

Errors are the path, not the obstacle. 404 -> Docker networking; 403 -> two kinds of protection working; exit 1 -> one bad character in a file. Read the last line of the error first.

Free tiers are real infrastructure: Cloudflare CDN + Mapbox 100k requests + OSM data + open-source Shareabouts = a launched product for about \$19/year + \$12/month.

Journal generated 18 July 2026 with Claude. kothakhaj.com - kotha khoj, sajilai.